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PULCHOWK CAMPUS

A

REPORT

ON

# TEXT CLASSIFICATION FOR DATA LOSS PREVENTION (DLP)

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# Abstract

In the era of digital transformation, protecting sensitive information has become increasingly critical due to the rising threat of data breaches, which can lead to significant financial loss and reputational damage for organizations. Traditional Data Loss Prevention (DLP) systems often rely on predefined rules that fail to adapt to the evolving nature of data leaks and sensitive information exfiltration. This report presents a machine learning-based approach for text classification aimed at enhancing the effectiveness of DLP systems. By employing advanced techniques such as Support Vector Machines (SVM) and Long Short-Term Memory (LSTM) networks, the proposed two-step classification model effectively distinguishes between enterprise-related and non-enterprise documents, and further categorizes sensitive versus non-sensitive data. The study evaluates model performance through various metrics, demonstrating high accuracy, recall, and precision, ultimately contributing to improved organizational data governance and compliance.

**Keywords:** Data Loss Prevention (DLP), Text Mining, Support Vector Machine (SVM), Long Short-Term Memory (LSTM), Sensitive Information, Natural Language Processing (NLP), Data Security, Cybersecurity.

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# List of Abbreviations

| | |
|---|---|
| **PII** | Personally Identifiable Information |
| **SPII** | Sensitive Personally Identifiable Information |
| **DLP** | Data Loss Prevention |
| **LSTM** | Long Short Term Memory |
| **SVM** | Support Vector Machine |
| **NLP** | Natural Language Processing |
| **RNN** | Recurrent Neural Network |
| **RBF** | Radial Basic Function |
| **TF-IDF** | Term Frequency - Inverse Document Frequency |

# 1. Introduction

In this digital transformation era, sensitive information security has become of utmost importance. Organizations in every field are being threatened by data breaches these days, resulting in potentially huge financial losses, reputational damage, and possible legal consequences. A DLP system is generally one kind of security technology meant for detecting and preventing the unauthorized transfer of sensitive data outside an organization's network. Meanwhile, with constantly growing volumes of textual data such as emails, documents, and instant messages, classic rule-based DLP systems are usually not effective enough. In this work, a machine learning approach is developed to implement text classification for improving the effectiveness of DLP systems in identifying and mitigating the risk of data loss.

## 1.1 Background

Traditional Data Loss Prevention systems work with a predefined set of rules in order to monitor and protect sensitive data, usually focusing on patterns such as credit card numbers, social security numbers, or other personally identifiable information. However, as the nature of data shifts, so too are the methods that malicious actors use to exfiltrate it. For example, employees may inadvertently leak sensitive information via channels that appear harmless, or the attacker may deliberately use techniques to cloak the data they are exfiltrating. Classic DLP solutions can have a hard time identifying complex, context-dependent leaks, resulting in either false negatives or superfluous false positives.

Recent advances in machine learning and NLP can help bridge this gap. A text classifier, following training on a large corpus of labeled data, can learn to identify patterns and context which rule-based systems often will miss. Text classification models can be tuned for the detection of a wide range of sensitive information types-from financial data and intellectual property to others-enhancing the overall accuracy and effectiveness of a DLP system.

## 1.2 Problem Statement

Many organizations rely on manual processes for classification and protection of sensitive information despite advancements in technology. The processes usually involve huge human effort, whereby employees or security personnel have to manually go through vast amounts of textual data, such as emails, documents, and chat logs, to identify potential data leaks. These methods are highly time-consuming and prone to errors. More importantly, this task

becomes increasingly difficult with an increase in the volume and complexity of the data. Also, traditional DLP solutions relying on predefined rules are basically limited to detecting context-dependent or obfuscated forms of sensitive information. Due to this, most organizations have been commonly experiencing high false-positive rates and missed threats. It deals with these challenges by developing a machine learning-based text classifier that will classify sensitive textual data automatically and, thus, reduce the workload of DLP systems manually. This also improves the accuracy and enhances the efficiency of any DLP system in preventing data loss.

## 1.3 Objectives

The main objective of the project is to design a text classifier that can categorize sensitive data in textual content. It will be implemented using different machine learning algorithms, and will be evaluated based on standard performance metrics such as accuracy, precision, recall, and F1 score.

In addition to the practical objective of building an effective classifier, this project also aims to contribute to the academic field of data security and machine learning.

## 1.4 Scope

- **Machine Learning-Based Text Classifier:** A machine learning model to classify textual data as sensitive versus non-sensitive, with a key emphasis on achieving the maximum accuracy, efficiency, and scalability in classification. The classifier will be trained and validated using a curated dataset of relevant textual data for ensuring robustness in the performance across a wide array of data types with minimal overfitting.

- **Automation and Data Security Enhancement:** It automates the classification of sensitive information, reducing the need for manual interference that can result in human errors and inconsistencies in data categorization. -The model will contribute to improving general data security by enhancing the precision and speed of the processes of detecting events that might potentially lead to data loss. Consequently, proactive and efficient strategies for data protection will be supported.

# 2. Literature Review

## 2.1 Related work

### 2.1.1 Text Classification and Natural Language Processing (NLP)

Text classification is a critical task in Natural Language Processing (NLP), and significant advancements have been made in this domain over the past few decades. The foundational work by Jurafsky and Martin [1] provides a comprehensive overview of text preprocessing techniques such as tokenization, stemming, lemmatization, and part-of-speech tagging, which are crucial for preparing textual data for classification tasks. In recent years, deep learning techniques have significantly improved the accuracy of text classification models. For instance, Kim [2] introduced a convolutional neural network (CNN) for sentence classification, demonstrating that deep learning models could outperform traditional machine learning methods when provided with sufficient data.

### 2.1.2 Support Vector Machines (SVM) for Text Classification

Support Vector Machines (SVM) have been a popular choice for text classification due to their effectiveness in high-dimensional spaces and their ability to handle sparse data, as is common in text data [3]. The seminal work by Cortes and Vapnik [4] laid the foundation for SVM, introducing the concept of maximizing the margin between data points and the decision boundary. Joachims [5] further extended the application of SVM to text classification, demonstrating its superiority over other traditional algorithms such as Naive Bayes and k-Nearest Neighbors, especially when combined with techniques like TF-IDF for feature representation.

### 2.1.3 Long Short-Term Memory (LSTM) Networks in Text Processing

Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN), have become the standard for sequence modeling tasks, including text classification. The work by Hochreiter and Schmidhuber [6] introduced LSTM as a solution to the vanishing gradient problem faced by traditional RNNs, enabling the modeling of long-term dependencies in sequences. Recent studies, such as by Liu et al. [7], have applied LSTM networks to sentiment analysis and other text classification tasks, showing their effectiveness in capturing

contextual information over longer sequences.

### 2.1.4 Data Loss Prevention (DLP) and Text Classification

The integration of text classification techniques into Data Loss Prevention (DLP) systems has become increasingly important as organizations seek to prevent data breaches by accurately identifying sensitive information within documents and communications. Current DLP systems, as discussed by Forcepoint [8], utilize machine learning models to analyze text and enforce security policies based on the classification results. The application of SVMs in DLP systems has been explored by studies such as those by Li et al. [9], who applied SVMs to identify and categorize sensitive information, demonstrating the feasibility and effectiveness of this approach in a real-world DLP context.

## 2.2 Related theory

### 2.2.1 Data Loss Prevention (DLP)

Data Loss Prevention (DLP) refers to a set of technologies and inspection techniques designed to classify, monitor, and protect sensitive information contained within objects such as files, emails, packets, applications, or data stores. These systems operate across three primary states of data: at rest (in storage), in use (during operations), and in transit (across a network). DLP systems are integral to preventing the unauthorized transmission of sensitive information outside an organization's network by employing a combination of content inspection methods, including keyword matching, pattern recognition, and contextual analysis.

The DLP workflow involves several critical steps. First, data within an organization's infrastructure is classified, assigning specific confidentiality levels (e.g., public, sensitive, confidential, or top secret) to every piece of data [10]. This classification process can be manual or automated, depending on the DLP solution. Next, security policies and rules are developed, specifying how the system should respond to events, such as sending alerts, blocking data transmissions, or revoking access rights. Once the system is configured, it continuously monitors data movement and investigates triggered alerts, distinguishing between incidents and false positives. As new data enters the system, the classification process repeats to ensure consistent protection.

DLP solutions also leverage machine learning algorithms and Natural Language Processing (NLP) techniques to enhance accuracy, reducing false positives and improving detection capabilities for sensitive data, such as Personally Identifiable Information (PII), financial records, and intellectual property. By implementing robust security rules and automation, DLP systems mitigate the risks of data breaches, regulatory non-compliance, and insider

threats, forming a critical layer in an organization's cybersecurity strategy [8, 10].

### 2.2.2 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a subfield of artificial intelligence that focuses on the interaction between computers and human language. NLP enables machines to process, understand, and generate human language in a way that is both meaningful and useful. In text classification tasks, NLP techniques like tokenization, stemming, lemmatization, and part-of-speech tagging are employed to preprocess and analyze text data. A commonly used mathematical representation in NLP is the Term Frequency-Inverse Document Frequency (TF-IDF), which evaluates the importance of a word in a document relative to a corpus. The TF-IDF score for a term $t$ in a document $d$ is calculated as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

where $\text{TF}(t, d)$ is the frequency of term $t$ in document $d$, and $\text{IDF}(t)$ is the inverse document frequency, defined as

$$\text{IDF}(t) = \log\left(\frac{N}{1 + n_t}\right),$$

with $N$ being the total number of documents and $n_t$ the number of documents containing $t$. This weighting scheme helps in distinguishing words that are common across the corpus from those that are specific to certain documents [1].

### 2.2.3 Support Vector Machines (SVM)

Support Vector Machines (SVM) is a powerful supervised learning algorithm used for classification tasks. SVM works by finding the optimal hyperplane that maximizes the margin between different classes in a feature space. Mathematically, given a training dataset $\{(x_i, y_i)\}_{i=1}^{n}$, where $x_i$ represents the feature vectors and $y_i$ the class labels, SVM aims to solve the optimization problem:

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w \cdot x_i + b) \geq 1, \quad \forall i$$

Here, $w$ is the weight vector, and $b$ is the bias term. The support vectors are the data points closest to the hyperplane and are critical in defining its position. For non-linearly separable data, SVM uses kernel functions, such as the Radial Basis Function (RBF) kernel, to transform the data into a higher-dimensional space where a linear separation is possible. The kernel trick allows SVM to operate in this higher-dimensional space without explicitly computing the coordinates [4].

### 2.2.4 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to capture long-term dependencies in sequential data. Unlike traditional RNNs, LSTMs effectively mitigate the vanishing gradient problem, making them suitable for tasks where the context of earlier information is crucial. The key to LSTM's success is its internal memory cell and the three gates: input, forget, and output gates. The mathematical representation for an LSTM cell at time step $t$ includes:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(forget gate)}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(input gate)}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(cell candidate)}$$

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \quad \text{(cell state)}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(output gate)}$$

$$h_t = o_t \times \tanh(C_t) \quad \text{(hidden state)}$$

Where $\sigma$ is the sigmoid function, tanh is the hyperbolic tangent function, and $W_f, W_i, W_C, W_o$ are weight matrices. The internal gating mechanisms of LSTM allow it to selectively retain or forget information, making it powerful for text sequences where the meaning of a word can depend heavily on previous words in the sequence [6].

# 3. Methodology

The steps taken to develop an effective Data Loss Prevention (DLP) classifier, from corpus construction to training a robust, multi-layered model. The goal is to build a model capable of accurately distinguishing sensitive (private) from non-sensitive (public) documents while minimizing false positives and negatives, ensuring reliable performance in real-world enterprise environments.

## 3.1 Building the Corpora

The classifier requires labeled datasets containing representative samples of both sensitive and public documents. Five distinct corpora were constructed to represent various types of organizations and industries, using publicly available sources of sensitive information from leaks or archives, alongside public information from the same sources.

- **DynCorp Corpus:**
  - *Sensitive Documents*: Leaked field manuals used by DynCorp operatives, accessed through WikiLeaks[11].
  - *Public Documents*: Web pages from the official DynCorp website providing publicly accessible information about the company's services[12].
- **Transcendental Meditation (TM) Corpus:**
  - *Sensitive Documents*: Leaked internal documents, such as workshop instructions, sourced from WikiLeaks[11].
  - *Public Documents*: Documents and web pages from TM-affiliated websites detailing practices and events for a general audience[13].
- **Mormon Corpus:**
  - *Sensitive Documents*: The internal handbook of the Church of Jesus Christ of Latter-Day Saints, split into smaller text segments, and other confidential documents sourced from WikiLeaks[11].
  - *Public Documents*: Publicly available religious teachings and community materials from the official LDS website[14].

- **Enron Corpus:**
  - *Sensitive Documents*: Internal, business-related emails from the Federal Energy Regulatory Commission's (FERC) investigation into Enron[15].
  - *Public Documents*: Archived web pages from the Enron public website, accessed via the Wayback Machine[16].
- **Google Corpus:**
  - *Sensitive Documents*: Software development posts by Google employees on internal topics, simulating private, technical discussions[17].
  - *Public Documents*: PR-related blog posts from Google intended for external communication[18].

To introduce further realism and handle non-enterprise (NE) content, an additional non-enterprise corpus was compiled:

- *Non-Enterprise (NE) Data*: Wikipedia articles[19], Brown Corpus samples[15], and Reuters news articles[15], representing a variety of general knowledge and news content.

## 3.2 Preprocessing

- **Sensitive Information Masking:** Replace personal data (e.g., phone numbers, credit card numbers, account numbers, expiry dates, and CVC codes) with generic placeholders like `PHONE_NUM`, `CREDIT_CARD`, etc., to ensure privacy.
- **Text Normalization:**
  - Convert the entire text to lowercase for uniformity.
  - Remove non-alphabetic characters, except for the placeholders, to eliminate punctuation and special symbols.
- **Tokenization:** Split the text into individual words (tokens) for easier analysis.
- **Stop Word Removal:** Remove common, non-essential words (e.g., *the*, *is*, *and*) that don't add significant meaning.
- **Stemming:** Reduce words to their root form (e.g., *running* becomes *run*) to treat word variations as the same entity.
- **Result:** Produce a clean, anonymized, and standardized text, ready for further analysis, with sensitive information protected and irrelevant noise minimized.

## 3.3 Feature Extraction

Feature extraction from text data involves transforming raw text into numerical features that can be used for analysis or machine learning. **TF-IDF (Term Frequency-Inverse Document Frequency)** is a common method for this purpose.

- **Data Consolidation:** Text data from multiple sources (e.g., enterprise private, public, and non-enterprise) is combined into a single dataset.
- **Tokenization & N-grams:** The text is broken down into unigrams (single words) and bigrams (pairs of consecutive words) to capture both individual terms and meaningful phrases.
- **Term Frequency (TF):** Measures how often a term appears in a document, highlighting its importance within that document.
- **Inverse Document Frequency (IDF):** Measures how rare a term is across all documents, giving more weight to unique terms and reducing the importance of common ones.
- **Feature Selection:** The top 20,000 terms (including n-grams) are selected based on their TF-IDF scores, excluding common stop words.
- **Feature Representation:** Each document is then represented as a vector of TF-IDF values, capturing the importance of terms within the dataset.

## 3.4 Model Training and Cross-Validation

- **Cross-Validation**: The model is trained and evaluated using 10-fold cross-validation across each corpus, calculating false positive and false negative rates within sensitive and public classes to ensure balanced accuracy.
- **Testing on NE Data**: The classifier is further evaluated on the NE corpus to measure its generalization ability, testing how well it avoids classifying unrelated, non-enterprise content as sensitive.

## 3.5 Performance Metrics and Tuning

Performance metrics such as accuracy, precision, recall, and F1 score are critical to evaluating and tuning the classifier:

- **Accuracy:** Accuracy measures the proportion of correctly classified instances (both true positives and true negatives) among all predictions. While a useful general metric, it may not always reflect the classifier's performance in imbalanced datasets.

- **Precision:** Precision measures the proportion of true positive predictions among all instances classified as positive. High precision reduces the risk of false positives, making it particularly important in applications where false positives are costly or undesirable.

- **Recall:** Recall evaluates the proportion of true positive predictions among all actual positive instances. It is crucial for ensuring that the classifier identifies as many relevant instances as possible, even if it leads to a higher number of false positives.

- **F1 Score:** The F1 score is the harmonic mean of precision and recall, offering a balanced metric that accounts for both false positives and false negatives. It is especially useful when there is a need to balance precision and recall, such as in scenarios with imbalanced classes.

The classifier is iteratively tuned by adjusting the decision thresholds and optimizing these metrics to achieve the best balance between precision and recall, ensuring practical and reliable performance.

# 4. System Design

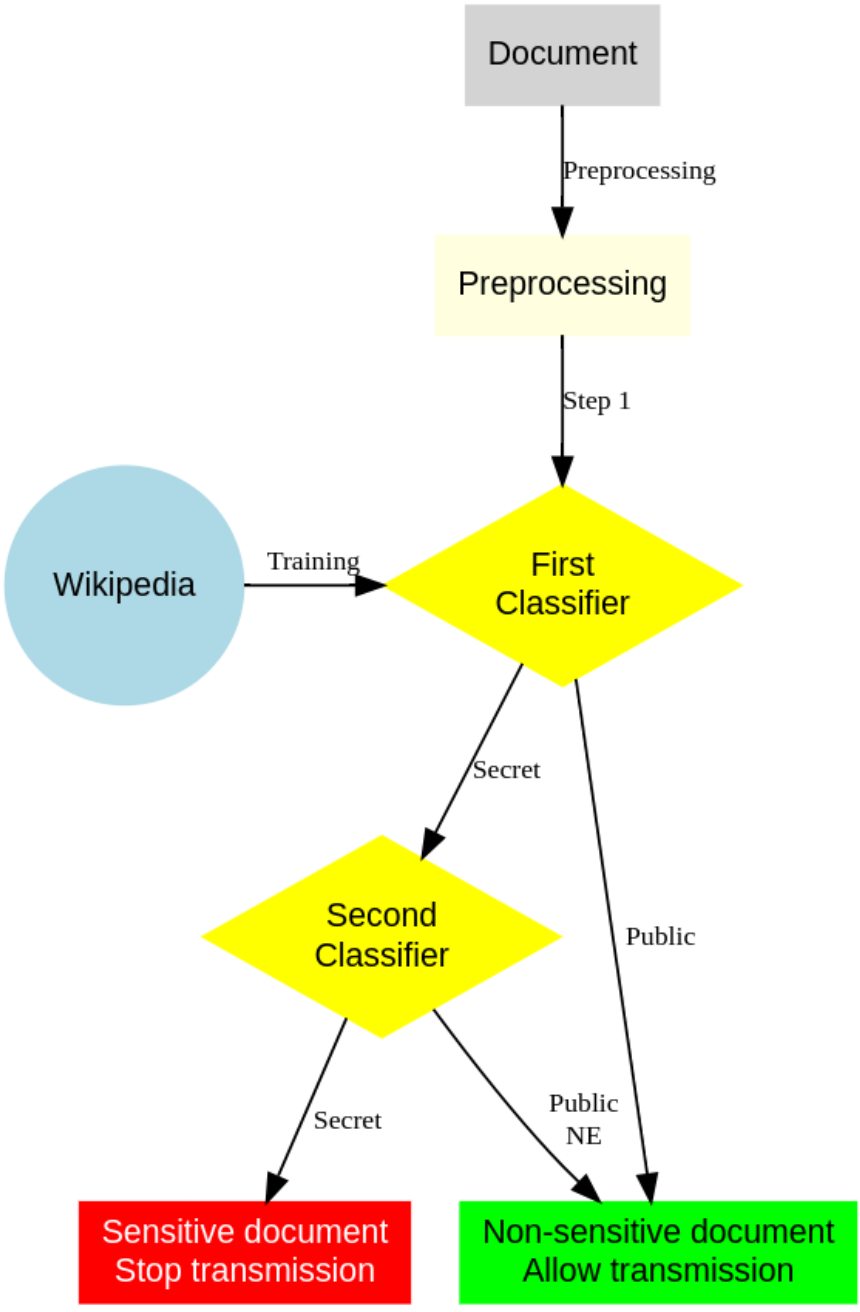



Figure 4.1: Two Step Classifier

## 4.1 Model Architecture: Two-Step Classification

The classification framework is structured as a two-step process to further refine detection accuracy.

### 4.1.1 Step 1: First Classifier (Enterprises vs Non-Enterprises)

In the first step, a classifier is trained to separate enterprise-related documents (both public and sensitive) from non-enterprise (NE) documents.

- *Objective*: The primary objective is to minimize false positives among NE documents. This classifier focuses on distinguishing between enterprise-related content and general, non-relevant content, ensuring that non-enterprise documents are excluded from further processing.

### 4.1.2 Step 2: Second Classifier (Sensitive vs Non-Sensitive)

For documents that have been classified as enterprise-related by the first classifier, a secondary classification step is performed to distinguish between enterprise public and enterprise-sensitive documents.

- *Objective*: The goal here is to classify enterprise documents into two categories—public and sensitive. This step ensures that sensitive documents are correctly identified and treated separately from public enterprise content.

## 4.2 Data Flow

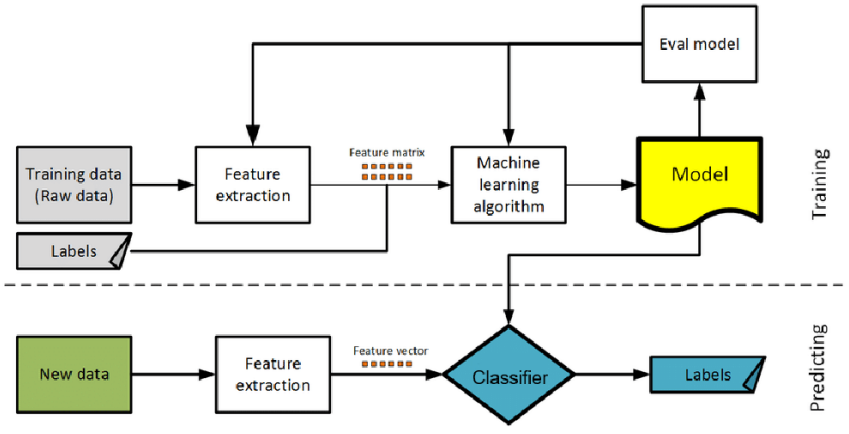



Figure 4.2: Data Flow Diagram

The data flow within the system is as follows:

- **Data Ingestion**: Data is collected from various sources and stored in a central repository.
- **Preprocessing**: Data is cleaned, tokenized, and features are extracted. Non-relevant data (e.g., stop words) is removed.
- **Feature Engineering**: The system applies techniques like TF-IDF or embeddings to transform the data into a suitable format for classification.
- **Classification Process**: During classification, documents are first passed through the SA Secret Classifier to identify if they belong to the enterprise or non-enterprise category. The second step uses Meta-Space Classification to determine whether the document is sensitive or not.
- **Results and Feedback**: The classification results are provided to the system's users (administrators, IT security teams), and feedback is incorporated for further tuning and continuous learning.

# 5. Result & Output

## 5.1 SVM Model

### 5.1.1 First Classifier

Table 5.1: Classifier Performance Report(SVM First Classifier)

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| 0 | 1.00 | 1.00 | 1.00 | 2288 |
| 1 | 1.00 | 1.00 | 1.00 | 1925 |
| **Accuracy** | 1.00 (4213 samples) | | | |
| **Macro Avg** | 1.00 | 1.00 | 1.00 | 4213 |
| **Weighted Avg** | 1.00 | 1.00 | 1.00 | 4213 |

### 5.1.2 Second Classifier

Table 5.2: Classifier Performance Report(SVM Second Classifier)

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| 0 | 0.98 | 0.96 | 0.97 | 136 |
| 1 | 1.00 | 1.00 | 1.00 | 1790 |
| **Accuracy** | 1.00 (1926 samples) | | | |
| **Macro Avg** | 0.99 | 0.98 | 0.99 | 1926 |
| **Weighted Avg** | 1.00 | 1.00 | 1.00 | 1926 |

## 5.2 LSTM Model

### 5.2.1 First Classifier

Table 5.3: Classifier Performance Report(LSTM First Classifier)

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| 0 | 0.98 | 0.99 | 0.99 | 2288 |
| 1 | 0.99 | 0.98 | 0.98 | 1925 |
| **Accuracy** | 0.98 (4213 samples) | | | |
| **Macro Avg** | 0.98 | 0.98 | 0.98 | 4213 |
| **Weighted Avg** | 0.98 | 0.98 | 0.98 | 4213 |

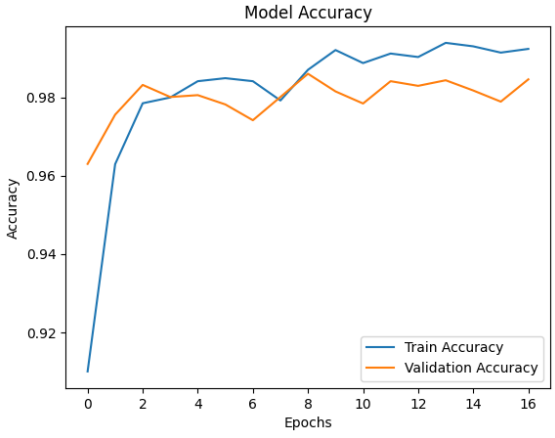



Figure 5.1: Step 1 LSTM Classifier Accuracy

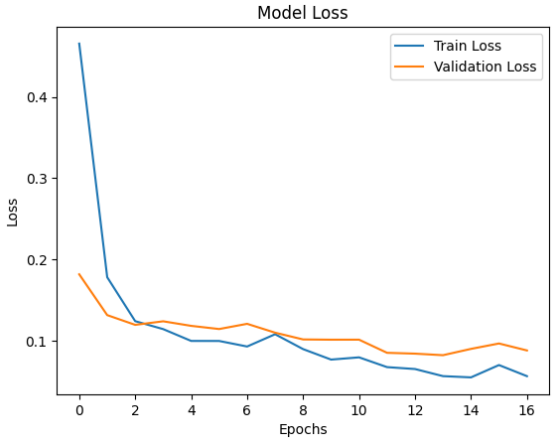



Figure 5.2: Step 1 LSTM Classifier Validation Loss

### 5.2.2 Second Classifier

Table 5.4: Classifier Performance Report(LSTM Second Classifier)

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| 0 | 0.92 | 0.96 | 0.94 | 136 |
| 1 | 1.00 | 0.99 | 1.00 | 1790 |
| **Accuracy** | 0.99 (1926 samples) | | | |
| **Macro Avg** | 0.96 | 0.98 | 0.97 | 1926 |
| **Weighted Avg** | 0.99 | 0.99 | 0.99 | 1926 |

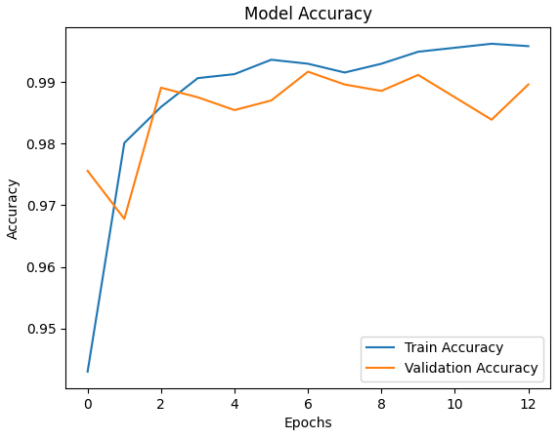



Figure 5.3: Step 2 Classifier Accuracy

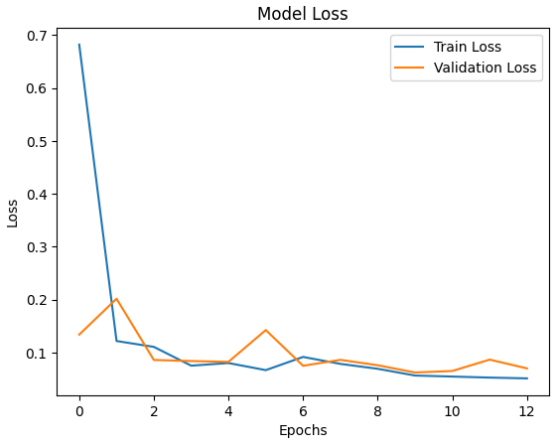



Figure 5.4: Step 2 Classifier Validation Loss

## 5.3 Performance Metrices on Validation Dataset

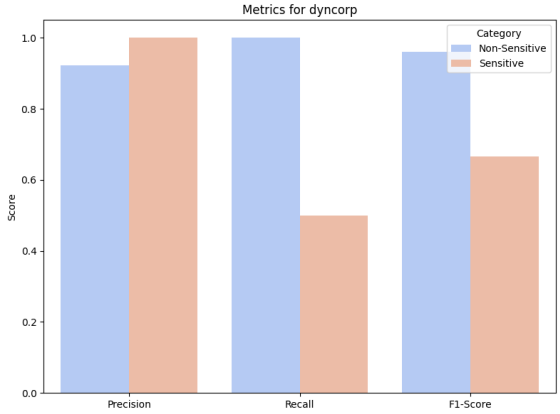



Figure 5.5: Performance Metrics for Dyn Corpus

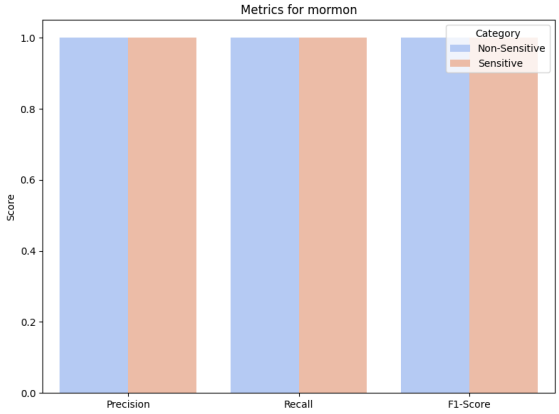



Figure 5.6: Performance Metrics for Mormon Corpus

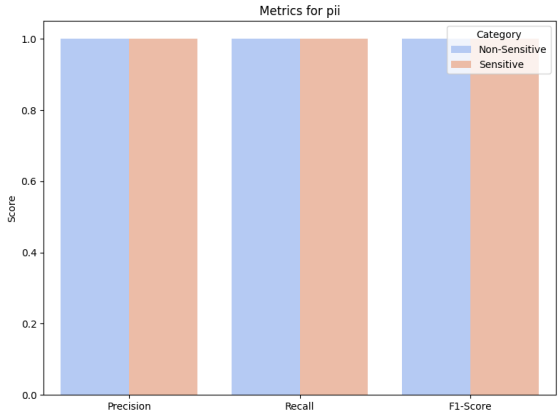



Figure 5.7: Performance Metrics for Pii Corpus

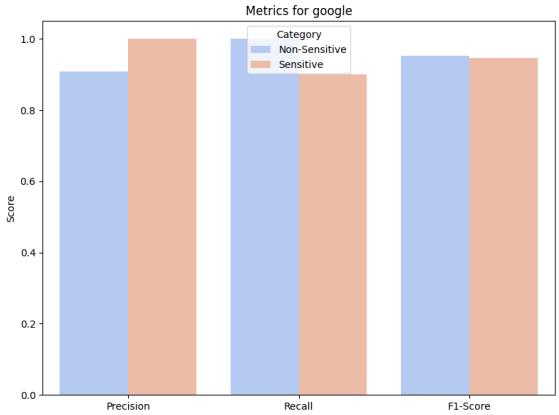



Figure 5.8: Performance Metrics for Google Corpus

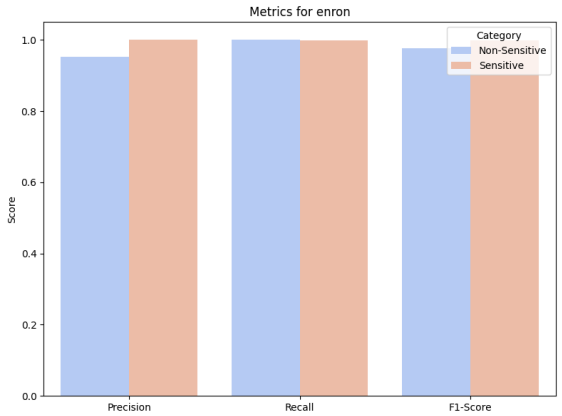



Figure 5.9: Performance Metrics for Enron Corpus

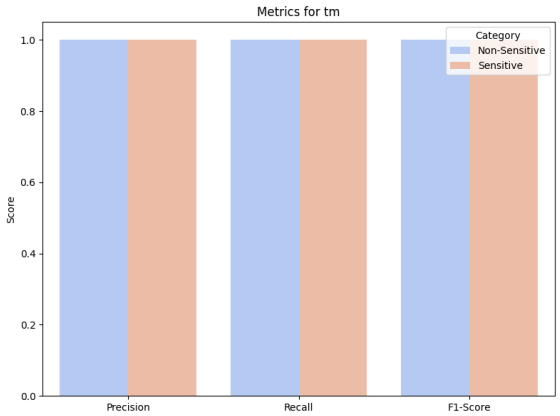



Figure 5.10: Performance Metrics for TM Corpus

## 5.4 Web Application Output

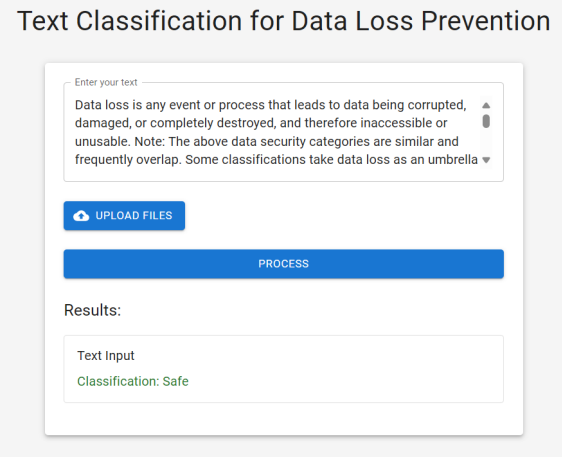



Figure 5.11: Two Step Classifier for text

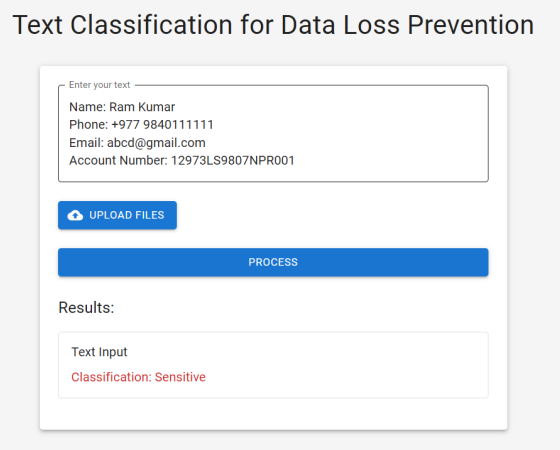



Figure 5.12: Two Step Classifier sensitive text

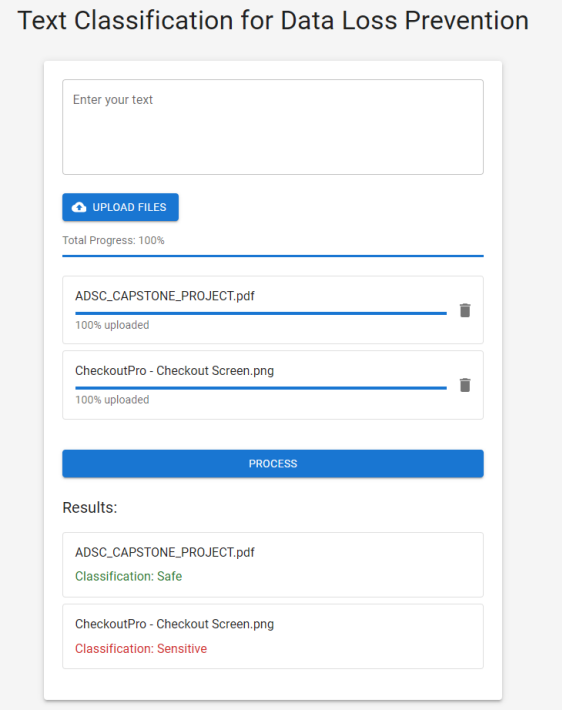



Figure 5.13: Two Step Classifier for file

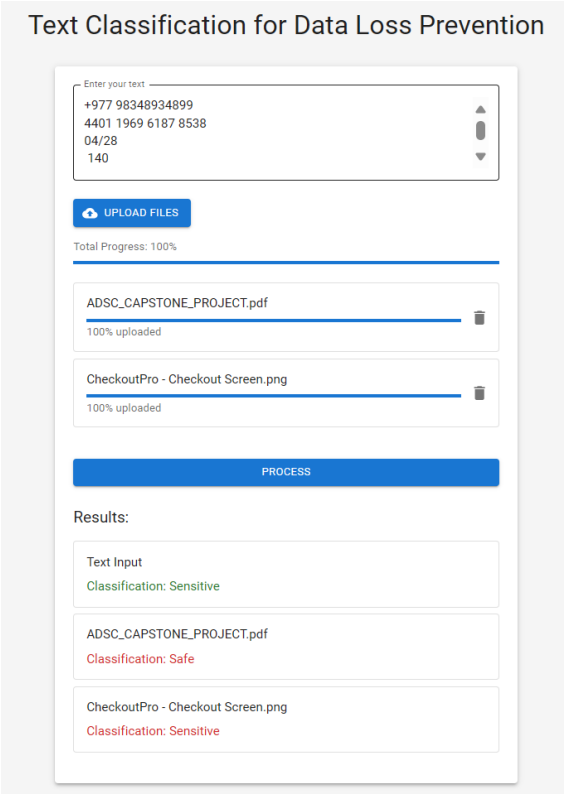



Figure 5.14: Two Step Classifier for both file and text

# 6. Epilogue

## 6.1 Use Cases

By automating the classification process, the system reduces the need for manual intervention, dramatically lowering the risk of errors and inconsistencies in data handling. This automation enhances operational efficiency while establishing a consistent and reliable framework for identifying and managing sensitive data. As a result, the system not only streamlines workflows but also reinforces organizational data governance, ensuring a higher standard of compliance and control over sensitive information.

## 6.2 Conclusion

In this study, we developed a two-step classifier using both traditional machine learning (SVM) and deep learning (LSTM) approaches, and evaluated their performances. The results demonstrated that both models achieved high accuracy, with the SVM classifier performing exceptionally well in terms of precision, recall, and F1-score, particularly in the first step of classification. The LSTM model, while slightly lagging behind in accuracy compared to SVM, still showed strong performance, particularly in the second step of the classification process.

The SVM model exhibited a high degree of efficiency, achieving perfect accuracy on the first classifier, indicating its strength in handling simpler, less complex tasks. On the other hand, the LSTM model was able to capture more intricate patterns, especially in the second classifier, demonstrating its capacity to handle more complex, sequential data. Although the LSTM model requires greater computational resources and time, its ability to model long-term dependencies makes it a valuable tool for more complex classification tasks.

Both classifiers, despite their differences, show great promise for real-world applications. The SVM model's speed and simplicity make it ideal for situations where quick deployment and efficiency are key, while the LSTM model is more suitable for tasks requiring the identification of complex, long-term relationships in data. The performance metrics suggest that further improvements in training time and model fine-tuning could lead to even better results for both approaches.

In conclusion, both traditional machine learning and deep learning approaches have their own advantages and can be applied to different real-world scenarios depending on the requirements of the task. The progress made within the constraints of the project timeframe

is promising, and both models show potential for real-world deployment in text classification tasks. Further research and refinement of these models could yield even better performance, particularly in more specialized domains.

## 6.3 Limitations

1. We have used a limited number of private leaked datasets from a few corporations, which may not fully represent the diversity of sensitive data across various industries.

2. Although we have created models for both traditional and deep learning approaches, we haven't combined them for testing whether one approach outperforms the other in a hybrid model configuration.

## 6.4 Further Enhancements

1. Integrating a broader range of leaked and sensitive datasets from various industries to improve the model's accuracy and robustness in detecting diverse sensitive information.

2. Incorporating advanced feature engineering techniques, such as semantic similarity and contextual embeddings, to enhance the system's ability to distinguish between nuanced sensitive and non-sensitive data.

3. Implementing a hybrid model that combines the strengths of both traditional machine learning and deep learning approaches to achieve a balance between accuracy and computational efficiency.

4. Developing continuous learning mechanisms that enable the model to dynamically adapt to new sensitive data patterns, ensuring its relevance and accuracy in evolving security contexts.

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